

Pedro Henrique Gesualdo Modesto

Physicist · MSc Researcher in Ultracold Atoms · Scientific Computing

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PROFILE

Physicist and MSc researcher at the São Carlos Institute of Physics, University of São Paulo, working on few-body quantum physics in ultracold atomic gases. I solve, write and review advanced physics problems and build numerical models that are tested against real experimental data. Experienced teaching assistant in Theoretical Mechanics, Statistical Physics and Computational Physics.

> **Top 10% nationally** in Brazil's Unified Graduate Physics Exam, 2026 (score 6.0/10, 90.6th percentile; national mean 4.1).

EDUCATION

MSc in Physics (Theoretical and Experimental Physics) 2026 – present
São Carlos Institute of Physics, University of São Paulo
Funded by a national graduate research fellowship (2026–2028).

BSc in Medical Physics 2019 – 2026
School of Philosophy, Sciences and Letters of Ribeirão Preto, University of São Paulo

Technical High School Diploma in Internet Computing 2016 – 2018
Etec Prof. Rodolpho José Del Guerra, São José do Rio Pardo

RESEARCH

MSc Research: Low-Energy Scattering in Ultracold Atoms 2026 – present
São Carlos Institute of Physics, University of São Paulo

- Numerical modeling of s-wave scattering, three-body recombination and Efimov physics in ultracold potassium-39.
- Direct comparison of models with three-body loss measurements from an experimental cold-atom laboratory.
- Reproducible research code in Python and Julia on a dedicated Linux workstation with automated tests.

Undergraduate Research: Computational Model of a Tripartite Synapse 2023 – 2024
Neural Systems Laboratory, University of São Paulo

- Python model of multicompartment neurons and neuron–astrocyte signaling, reproducing Durstewitz et al. (2000).
- Numerical simulation of dynamical systems and parameter analysis.
- Poster at the 32nd University of São Paulo International Symposium of Scientific and Technological Initiation (2024).

TEACHING

Teaching Assistant, Computational Physics 2025 – 2026
Programming, numerical methods and simulation of physical systems.

Teaching Assistant, Statistical Physics 2025
Statistical ensembles, Boltzmann distribution, partition functions, thermodynamic relations.

Teaching Assistant, Theoretical Mechanics 2025
Lagrangian and Hamiltonian formalisms, constraints, symmetries and conservation laws.

University Scholarship Project, Teaching of Electromagnetism 2023 – 2024
Methods and materials for undergraduate Electricity and Magnetism.

University Scholarship Project, Science Communication 2021 – 2022
Videos and articles presenting the school's research to the public.

PROFESSIONAL EXPERIENCE

Research Data Assistant, Oncology Genomic Cohort Project 2022 – 2023
Ribeirão Preto Medical School, University of São Paulo · 30 h/week

- Participant recruitment under research protocols; collection, organization and validation of clinical research data in a multidisciplinary team.

SKILLS

Physics: Classical and analytical mechanics, quantum mechanics, scattering theory, statistical physics, electromagnetism, atomic physics, mathematical methods

Computing: Python (NumPy, SciPy), Julia, C/C++, numerical methods, simulation of dynamical systems, Linux, Git, automated testing

Writing: LaTeX, rigorous step-by-step solutions, technical review and error-finding, science communication

Languages: Portuguese (native), English (professional working proficiency), Spanish (basic), French (basic)